Quantum Mechanics I — Exercise Sheet 11

Part of a pair

Variant: Questions only

Conventions, as in Lecture 11: a pure state of the pair has the coefficient matrix C, and its parts have $\rho_1 = CC^\dagger$ and $\rho_2 = C^{\mathsf{T}}\overline{C}$ (11.2); the probe of Example 11.11 has $K_0 = \operatorname{diag}(1, \cos(\theta/2))$ and $K_1 = \operatorname{diag}(0, \sin(\theta/2))$.

Exercise 1 (Schmidt's form and the entropy).

(a) For $\psi = \frac{1}{\sqrt{3}}(|00\rangle + |01\rangle + |10\rangle)$, find ρ_1, ρ_2, their eigenvalues and the entanglement entropy.

(b) For $\psi = \frac{1}{\sqrt{3}}(e_0 \otimes e_0 + e_1 \otimes e_1 + e_1 \otimes e_2)$ in $\mathbb{C}^2 \otimes \mathbb{C}^3$, write ψ in Schmidt's form (Theorem 11.4), find ρ_2 and the entanglement entropy.

(c) In the flip-flop of Lecture 10, from $|01\rangle$, the coefficient matrix at time t is, up to a phase, $\left(\begin{smallmatrix} 0 & (1+e^{\mathrm{i}\nu t})/2 \\ (1-e^{\mathrm{i}\nu t})/2 & 0 \end{smallmatrix}\right)$. Find $\rho_1(t)$ and its entropy. When is the entropy largest, and when is it 0?

Exercise 2 (the thermal state has the largest entropy). Let H be a Hamiltonian on $\mathbb{C}^n$, $\beta > 0$, $Z(\beta) = \operatorname{Tr} e^{-\beta H}$, and let $\rho = \sum_j q_j |v_j\rangle\langle v_j|$ be a density matrix with an orthonormal eigenbasis $v_1, \dots, v_n$.

(a) Show that $e^y \geq 1 + y$ for real y, with equality only at $y = 0$.

(b) Let v be a unit vector. Show that $e^{-\beta\langle v, Hv\rangle} \leq \langle v, e^{-\beta H} v\rangle$, with equality exactly when v lies in one eigenspace of H.

(c) Put $a_j = e^{-\beta\langle v_j, Hv_j\rangle}$, $I = \{j : q_j > 0\}$ and $A = \sum_{j\in I} a_j$. Show that $\sum_{j\in I} q_j \log(a_j/q_j) \leq \log A$, with equality exactly when $q_j = a_j/A$ for every $j \in I$.

(d) Conclude that $S(\rho) - \beta \operatorname{Tr}(\rho H) \leq \log Z(\beta)$, with equality only for $\rho = \rho_\beta$ of (11.3).

(e) Show that among all density matrices with the mean energy $\operatorname{Tr}(\rho_\beta H)$, the thermal state alone has the largest entropy.

Exercise 3 (the thermal state of T). The NV centre's spin of Lecture 4 has $H_T = \varepsilon Z^2$ in the basis e_+, e_0, e_-, with $\varepsilon/h = D = 2.87$ GHz.

(a) Find ρ_β and the probability of the reading $m_s = 0$. Evaluate $\beta\varepsilon = hD/(k_B T)$ and this probability at 300 K and at 4 K.

(b) A reading of the energy gives ε. In what state is the spin then? The state $|\chi_+\rangle\langle\chi_+|$, $\chi_+ = \frac{1}{\sqrt{2}}(e_+ + e_-)$, is stationary and lies in the same level; why is it not the answer?

(c) What does ρ_β tend to as $\beta \to 0$, and as $\beta \to \infty$?

Exercise 4 (the transpose is a reflection). A qubit has $\rho = \frac{1}{2}(\mathbb{1} + \mathbf{r}\cdot\boldsymbol{\sigma})$. Show that the second part of its purification, $\rho_2 = \rho^{\mathsf{T}}$ (Theorem 11.10), has the Bloch vector $(r_x, -r_y, r_z)$. Check it on $\rho = |+y\rangle\langle+y|$, with $|\pm y\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm \mathrm{i}\,|1\rangle)$.

Exercise 5 (the probe, unrecorded). In Example 11.11 the probe's reading is made and not recorded. Show that a qubit with the Bloch vector $\mathbf{r}$ is left with

$$\mathbf{r}' = \big(r_x \cos(\theta/2),\, r_y \cos(\theta/2),\, r_z\big).$$

What does $\theta = \pi$ give? What do n such probes in a row do, for $0 < \theta < \pi$? Now let the reading be recorded, and let $0 < \theta < \pi$. Show that the state left by the reading 0 is not given by Lüders' rule for any projection E. With what probability does the probe fire (read 1) on $\psi_0 |0\rangle + \psi_1 |1\rangle$, and what does its silence leave?

Exercise 6 (⋆ purifications differ by a unitary on the second system).

(a) Let C and D be coefficient matrices of two pure states of $\mathbb{C}^n \otimes \mathbb{C}^n$ with the same ρ_1, and let ρ_1 be invertible. Show that $V = C^{-1}D$ is unitary and that the second state is $(\mathbb{1} \otimes V^{\mathsf{T}})$ applied to the first.

(b) Find the unitary that takes $\Phi^+ = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ to ψ^-.